

HSE 712 Foundations of Biostatistics

Two-sample t test

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 - Hypothesis testing for the difference in means
 - Confidence intervals for the difference in means

$$b > \frac{1}{n} \sum_{i=1}^n x_i$$

Summary of t procedures for a sample mean

One-sample t test ✓	Paired/matched t test ✓	Two-sample t test Now  DARTMOUTH
When - To compare the mean of one sampled population to a hypothesized value (e.g., a reference population) σ^2 is unknown Dependent Variable - A continuous random variable, X Assumptions - Data are from a simple random sample X is normally distributed X is i.i.d.	When - To compare the mean within-pair difference to a hypothesized value (e.g., to zero) σ^2 is unknown Dependent Variable - The within-pair change in a continuous random variable, ΔX Assumptions - Data are from a simple random sample ΔX is normally distributed ΔX is i.i.d.	When - To compare the mean of two sampled populations to each other σ^2 is unknown Dependent Variable - The same continuous random variable measured in two populations (X_1, X_2) Assumptions - Data are from a simple random sample - Both X_1, X_2 are normally distributed - Both X_1, X_2 are each i.i.d.
i.i.d.: Identically and independently distributed. Each value for a participant is from the same distribution, and each participant is independent from each other.		

Two-sample t test

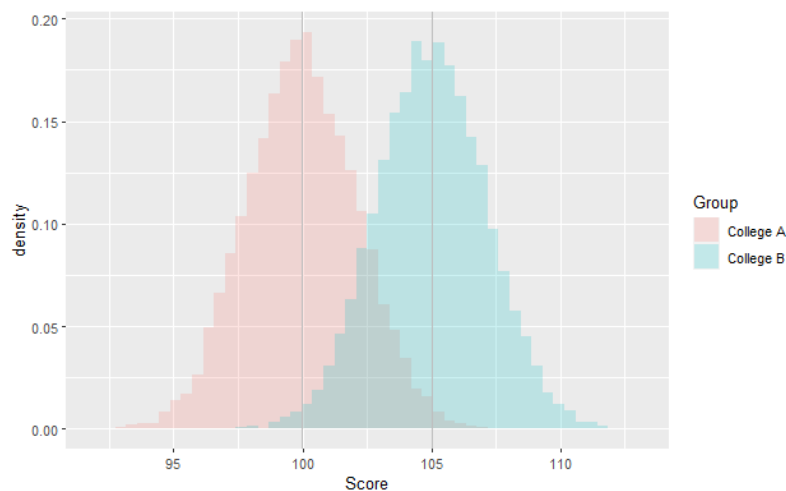
- When
 - To compare the means of two sampled populations to each other
 - Each σ^2 is unknown and are estimated by the sample variances
- Dependent Variable
 - A continuous random variable measured in two populations (X_1, X_2)
- Assumptions
 - Data are from two random samples
 - Both X_1, X_2 are normally distributed (or close to normal)
 - Both X_1, X_2 are each i.i.d.

Two-sample t test

- A two-sample t test is used to compare the expected means of two populations, as estimated from two random samples, to each other.
 - Are the random samples from the same underlying population with respect to the random variable? Do the two populations have the same sampling distributions?
- The variance (σ^2) in each population is unknown.
- The variances (σ^2) in the two populations can be assumed to be equal or unequal.

Example

- Suppose X is a random variable that follows a normal distribution. X is measured on 50 participants each in two groups sampled from College A and College B.
- We observed
 - $\bar{x}_A = 100, s_A = 15$
 - $\bar{x}_B = 105, s_B = 12$
- Here is a plot of the two sampling distributions on the same axis:



- Notice that if College A was the true sampling distribution, a sample mean of 105 for College B would be very rare (a mean of 105 is in the far right tail of the sampling distribution for College A.)
- Likewise, if College B was the true sampling distribution, a sample mean of 100 for College A would be very rare (a mean of 100 is in the far left tail of the sampling distribution for College B.)

Hypothesis testing for a two-sample t test

- A two-sample t test tests if the difference in sample means is zero or not.

- Null hypothesis: $H_o: \mu_A = \mu_B$
Equivalent to $H_o: \mu_A - \mu_B = 0$

- Alternative hypothesis:
 $H_o: \mu_A \neq \mu_B$ (two-sided alternative)
or
 $H_o: \mu_A > \mu_B$ (one-sided alternative)
or
 $H_o: \mu_A < \mu_B$ (one-sided alternative)

- Set $\alpha = 0.05$

- We know the sampling distribution for each sampled population:

$\bar{x}_A \sim N(\mu_{X_A}, \text{var} = \frac{\sigma_{X_A}^2}{n_A})$, approximated by $\bar{x}_A \sim t(\mu_{X_A}, \text{var} = \frac{s_{X_A}^2}{n_A})$ when $\sigma_{X_A}^2$ is unknown

$\bar{x}_B \sim N(\mu_{X_B}, \text{var} = \frac{\sigma_{X_B}^2}{n_B})$, approximated by $\bar{x}_B \sim t(\mu_{X_B}, \text{var} = \frac{s_{X_B}^2}{n_B})$ when $\sigma_{X_B}^2$ is unknown

Since we want to test $H_o: \mu_A - \mu_B = 0$, we must determine the sampling distribution of $\bar{x}_A - \bar{x}_B$.

- ➔ The sampling distribution of the difference in two sample means needs to be defined in order to derive the two-sample t test statistic. We will leverage rules of linear combinations to define that distribution.

Defining the sampling distribution for the difference in two sample means

- Let L be a new variable that is a linear combination of n independent random variables
(Each X_i is a different random variable, and there are n of them)

$$L = \sum_{i=1}^n c_i X_i = c_1 X_1 + c_2 X_2 + \cdots c_n X_n$$

- For a linear combination L , of n independent random variables, the following is true

$$E(L) = \sum_{i=1}^n c_i E(X_i)$$

$$Var(L) = \sum_{i=1}^n c_i^2 Var(X_i)$$

- Why?

See the handout, *A Brief Review of $E(X)$ and $Var(X)$* , for the rules needed to complete these solutions.

$$\begin{aligned} E(L) &= E(c_1 X_1 + c_2 X_2 + \cdots + c_n X_n) \\ &= E(c_1 * X_1) + E(c_2 * X_2) + \cdots + E(c_n X_n) \\ &= c_1 E(X_1) + c_2 E(X_2) + \cdots + c_n E(X_n) \\ &= \sum_{i=1}^n c_i E(X_i) \end{aligned}$$

$$\begin{aligned} Var(L) &= Var(c_1 X_1 + c_2 X_2 + \cdots + c_n X_n) \\ &= Var(c_1 X_1) + Var(c_2 X_2) + \cdots + Var(c_n X_n) \\ &= c_1^2 Var(X_1) + c_2^2 Var(X_1) + \cdots + c_n^2 Var(X_n) \\ &= \sum_{i=1}^n c_i^2 Var(X_i) \end{aligned}$$

Each of these random variables are independent of each other, and the covariance between any two random variables is zero. The covariance terms thus drop out of the equation for $Var(L)$.

- For the two-sample t test of means, we want to compare $\bar{x}_A - \bar{x}_B$. Let $L = (1)(\bar{x}_A) + (-1)(\bar{x}_B)$ (rewritten as a linear combination). Since $E(L) = \sum_{i=1}^n c_i E(X_i)$ in the general case, $E(L) = \mu_{X_A} - \mu_{X_B}$ for this case. Since $Var(L) = \sum_{i=1}^n c_i^2 Var(X_i)$ in the general case, $Var(L) = (1)^2 * \frac{\sigma_{X_A}^2}{n_A} + (-1)^2 * \frac{\sigma_{X_B}^2}{n_B} = \frac{\sigma_{X_A}^2}{n_A} + \frac{\sigma_{X_B}^2}{n_B}$ for this case and is approximated by $\widehat{Var}(L) = \frac{s_{X_A}^2}{n_A} + \frac{s_{X_B}^2}{n_B}$ when $\sigma_{X_A}^2$ and $\sigma_{X_B}^2$ are unknown. Since a linear combination of two (close to) normally distributed variables follows a normal distribution, and since we are approximating the true variances with sample variances, the difference in two means follows a t distribution:

The two-sample t test for independent samples

- The difference in two means from two normally distributed sampling distributions (here denoted by 1 and 2) is a linear combination and is normally distributed with mean

$$(\bar{x}_1 - \bar{x}_2) \text{ and variance } \frac{\sigma_{X_A}^2}{n_A} + \frac{\sigma_{X_B}^2}{n_B}.$$

$$(\bar{x}_1 - \bar{x}_2) \sim N[\mu_1 - \mu_2, \left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \right)]$$

- When we don't know the true variances of each variable, we estimate it with our observed data, and the difference in two means follows a t distribution.

$$(\bar{x}_1 - \bar{x}_2) \sim t_{df}[\mu_1 - \mu_2, \left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)]$$

- The deviation from the expected mean difference under the null can be standardized to a t score.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- If it is assumed that variances in the two populations are equal:

- Use a pooled variance, $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$ and the degrees of freedom are:

$df = n_1 + n_2 - 2$ where n_1 and n_2 are the sample sizes for the two samples.

- The test statistic simplifies to

$$t_{df=(n_1+n_2-2)} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

- If it is **not** assumed that variances in the two populations are equal:

- Use the two sample variances to estimate the population variances.
- The degrees of freedom are computed via **Satterthwaite's Method**:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\left(\frac{1}{n_1-1}\right) * \left(\frac{s_1^2}{n_1}\right)^2 + \left(\frac{1}{n_2-1}\right) * \left(\frac{s_2^2}{n_2}\right)^2}$$

- ...and the t test statistic remains:

$$t_{df=(see\ above)} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- This between group t test procedure with unequal variances is called Welch's t test.
- This is the **default** method for R's t.test()

Equal variances or not: Why choose one method over the other?
--

- The version that assumes equal variances is simpler and traditionally easier to compute.
- The version that assumes equal variances can have slightly greater power when the variances are truly the same in the populations.
 - This is most evident for small sample sizes, and this may not be a concern for moderate to larger sample sizes.
 - That gain in power depends on the degree of imbalance in the study.
- Whichever method you want to use – make sure to specify a priori.
- There are statistical procedures to test the equal variance assumption (coming up next).

Confidence intervals for the difference in two group means

- A two-sided 95% CI for the difference in means, assuming equal variances in the two groups:

$$(\bar{x}_A - \bar{x}_B) \pm t_{df=n_1+n_2-2, 1-\alpha/2}^* \cdot s_p \cdot \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

...where t^* is the critical value for the specified alpha level

- A two-sided 95% CI for the difference in means, assuming unequal variances in the two groups:

$$(\bar{x}_A - \bar{x}_B) \pm t_{df=See\ below, 1-\alpha/2}^* \cdot \sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}$$

...where t^* is the critical value for the specified alpha level and degrees of freedom are computed using the Satterthwaite's Method:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\left(\frac{1}{n_1-1}\right) \cdot \left(\frac{s_1^2}{n_1}\right)^2 + \left(\frac{1}{n_2-1}\right) \cdot \left(\frac{s_2^2}{n_2}\right)^2}$$

Notice that the generic form for a confidence interval around a summary statistic is:

$(1 - \alpha)\% \text{ CI} = \text{summary statistic} \pm \text{Critical Value} \cdot \text{standard deviation of the statistic}$

Example

This example is adapted from Rosner, problem 8.49.

A study was conducted of genetic and environmental influences on cholesterol levels. The data set used for the study were obtained from a twin registry in Sweden [10]. Specifically, four populations of adult twins were studied: (1) monozygotic (MZ) twins reared apart, (2) MZ twins reared together, (3) dizygotic (DZ) twins reared apart, and (4) DZ twins reared together. One issue is whether it is necessary to correct for gender before performing more complex genetic analyses. The data in Table 8.17 were presented for total cholesterol levels for MZ twins reared apart, by gender. The stratum for men includes 22 pairs of twins and thus 44 individuals in total. The stratum for women includes 24 pairs of twins and thus 48 individuals in total.

TABLE 8.17 Comparison of mean total cholesterol for MZ twins reared apart, by gender

	Men	Women
Mean	253.3	271.0
<i>sd</i>	44.1	44.1
<i>n</i>	44	48

Note: *n* = number of people (e.g., for males, 22 pairs of twins = 44 people) the cholesterol levels within each sample consist of independent observation.

If we assume (a) serum cholesterol is normally distributed, (b) the cholesterol levels within each sample consist of *independent* observations, and (c) the standard deviations for men and women are *not* the same, complete the statistical procedure to compare the two groups.

- Set up the research question
 - Is mean serum cholesterol the same or different between male and female monozygotic twins reared apart?
- Set up the research hypothesis
 - Mean serum cholesterol is the same between male and female monozygotic twins reared apart.
 - Notice no specific hypothesis or statement about the assumed direction of any difference by gender was provided, just that the researchers wanted to know if there was a difference by gender. There researchers may have also a priori assumed cholesterol concentrations would be greater for males than females. Regardless, the research question and statistical hypothesis would be the same (yes, the use of a two-sided alternative hypothesis is best even if the research hypothesis supposes a direction of the between-group difference).
- Define the test to use
 - Two-sample t test to compare mean values between two groups with unequal variances (Welch's t test)
- Are the assumptions of the approach valid?

- Yes
 - The outcome variable of interest, total serum cholesterol, is assumed to be sampled from a normal distribution.
 - The individuals within groups are assumed to be independent of each other per the note below Table 8.17.
 - Additionally, we are told to not assume equal variances and thus the approach to use is a two-sample t test with unequal variances, or Welch's t test.
- Set up the statistical hypotheses to test
 - Null hypothesis (H_0): Mean total serum cholesterol is the same between male and female monozygotic twins reared apart.
 - $H_0: \mu_M = \mu_F$ a.k.a., $\mu_M - \mu_F = 0$
 - Alternative hypothesis (H_A): Mean total serum cholesterol is not the same between male and female monozygotic twins reared apart.
 - $H_A: \mu_M \neq \mu_F$ a.k.a., $\mu_M - \mu_F \neq 0$
 - $\alpha = 0.05$
- Complete the test
 - The sample estimate for the difference in the two means is $\bar{x}_M - \bar{x}_F$, which is computed as a between-group difference of 17.7, with males having a lower mean.
 - That estimate follows a t distribution with $df = 89.3$ as computed using Satterthwaite's approximation, with mean and variance:

$$(\bar{x}_M - \bar{x}_F) \sim t_{89.3} \left(\mu_M - \mu_F, \frac{s_M^2}{n_M} + \frac{s_F^2}{n_F} \right)$$
 - That sampling distribution can be standardized with a mean of 0 and SE=1.
 - The t test statistic comparing the observed difference in mean is thus:

$$t = \frac{(253.3 - 271.0) - (0)}{\sqrt{\frac{44.1^2}{44} + \frac{44.1^2}{48}}} = -1.9230$$
 - The t test statistic is <0, and the final p-value is then
 - $2 * P(t \leq -1.9230) \approx 2 * 0.0288 \approx .0577$
 - The final p-value is not <0.05, and thus the null hypothesis is not rejected. There is not enough evidence to conclude that mean total serum cholesterol concentrations are difference between male and female monozygotic twins reared apart.
- Notice this final p-value is close to 0.05. I personally would state that the null is rejected to be consistent with the a priori research plan, yet the sample sizes are small and a difference of 17.7 might be clinically important (I don't know....). Thus, I might state that more research with larger sample sizes is needed.

Let's simulate the data to compute this test in R. Remember, the results will differ from those above because the data are being simulated.

Student's t-Test

Description

Performs one and two sample t-tests on vectors of data.

Usage

```
t.test(x, ...)  
  
## Default S3 method:  
t.test(x, y = NULL,  
       alternative = c("two.sided", "less", "greater"),  
       mu = 0, paired = FALSE, var.equal = FALSE,  
       conf.level = 0.95, ...)
```

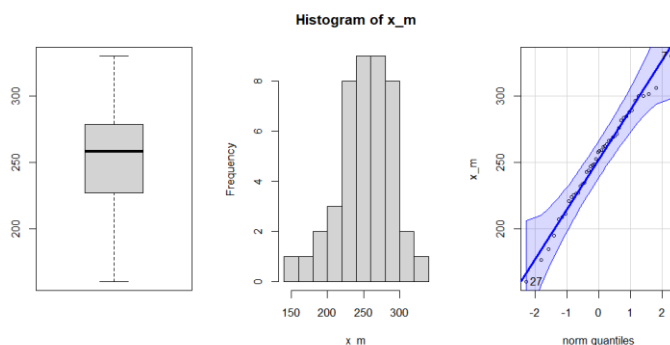
```
# I will set a seed so I can replicate my findings  
set.seed(05262025)
```

```
# library for the QQ Plot  
library(car)
```

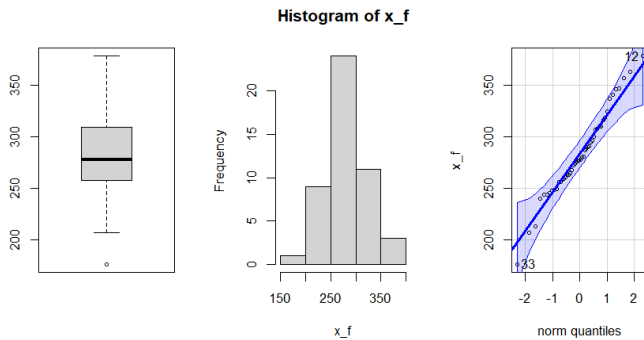
```
# Simulate the two variables, both from a normal distribution  
# Use the values from the example as the true mean and variances
```

```
x_m <- rnorm(44, 253.3, 44.1)  
x_f <- rnorm(48, 271.0, 44.1)  
# Remember rnorm requires n, mu, and sd (not variance)
```

```
# Summarize each  
par(mfrow=c(1,3))  
boxplot(x_m)  
hist(x_m)  
qqPlot(x_m)
```

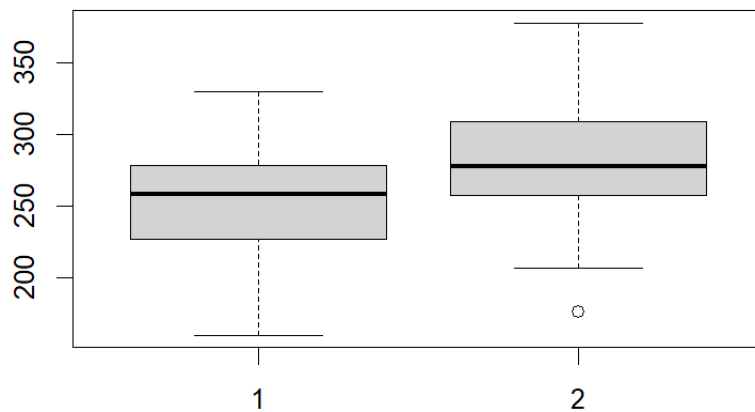


```
par(mfrow=c(1,3))
boxplot(x_f)
hist(x_f)
qqPlot(x_f)
```



No reason to suspect either variable was sampled from a NON-normal distribution.

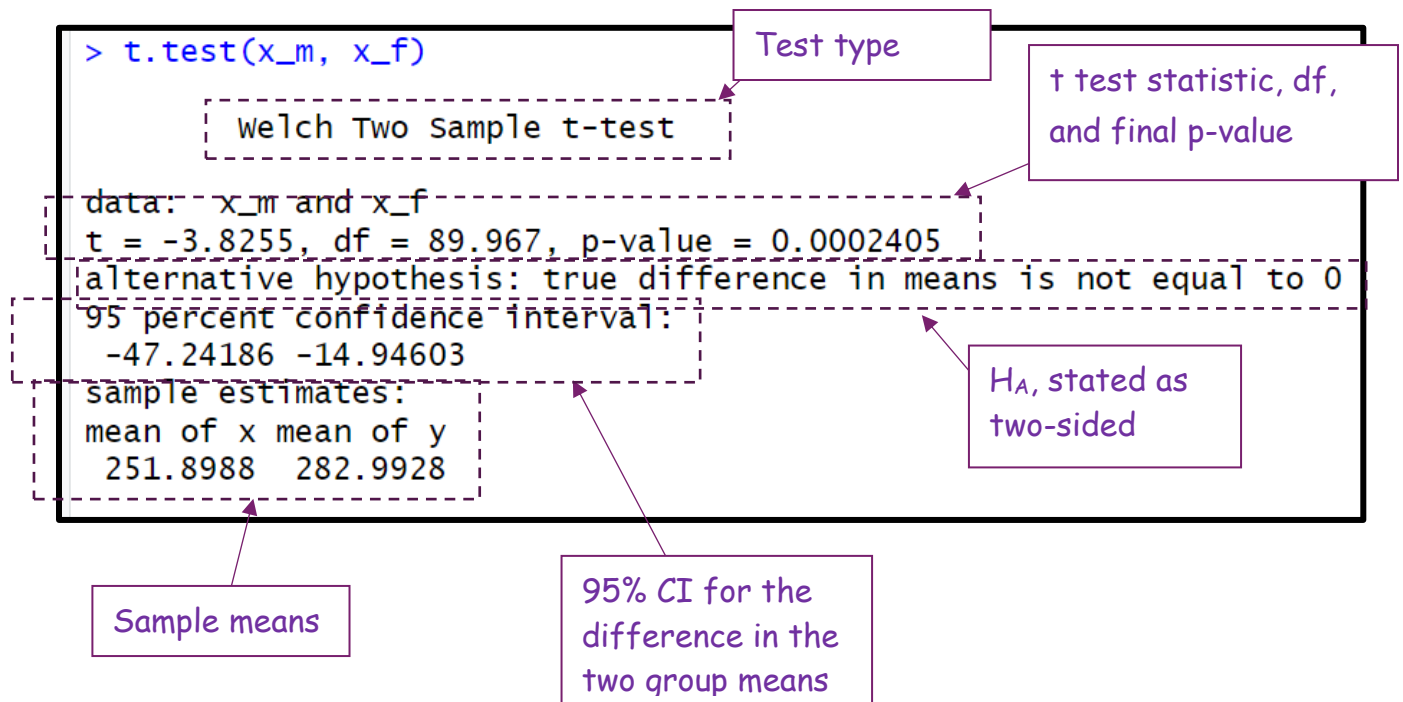
```
# Direct visual comparison by gender
dev.off() # Reset plotting window
boxplot(x_m,x_f)
```



Men on the left (1) and Women on the right (2). The distribution indeed appears shift rightward (toward larger values) for women.

```
# t test with unequal variances, expected difference in means = 0:
t.test(x_m, x_f, mu=0, conf.level = 0.95, alternative = "two.sided",
var.equal = FALSE)
```

```
# Same as (because of default settings):
t.test(x_m, x_f)
```



In the above example, it was assumed that the individuals within each stratum of gender (males or females) were independent of each other. In reality, that assumption is NOT valid. Why? Because pairs of twins were within each stratum, and those pairs are not independent because they shared the same genetic makeup. Because genetics can influence the outcome, this testing approach is not valid (but it's a useful exercise...). In reality, a testing procedure that can account for the clustering within groups is needed, such as a hierarchical model or mixed effects model.